Algorithms

Week 2 – Algorithm Design and Complexity Analysis

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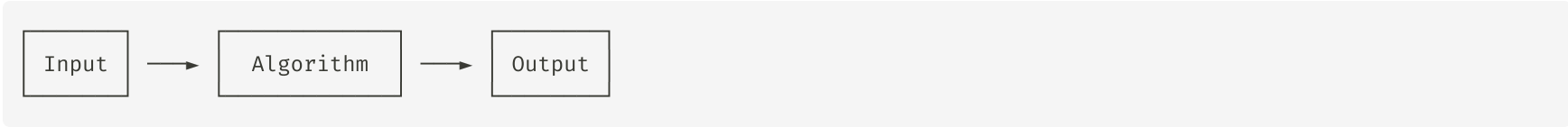
Part 1. Algorithm Fundamentals

Review: What is an Algorithm?

- An **algorithm** is a step-by-step procedure or method for solving a problem
- The problems we consider are solved **using a computer**
- An algorithm takes **input** and produces **output** (a solution)



Properties of Algorithms

Property	Description
Correctness	Must produce the correct output for every valid input (same input → same output)
Executability	Each step must be executable on a computer
Finiteness	Must terminate within a finite amount of time
Efficiency	The more efficient (in time and space), the more valuable

Euclid's GCD Algorithm

- The **oldest known algorithm**, dating back to ~300 BC
- **Greatest Common Divisor (GCD)**: the largest number that divides two natural numbers
- Key insight: GCD(A, B) = GCD(B, A mod B)

Mathematical basis:

If $A > B$, $A = a \cdot G$, $B = b \cdot G$ (where G = GCD), then:

$$A \mod B = (a \mod b) \cdot G$$

Repeatedly applying this reduces B to 0, at which point A is the GCD.

Euclid's GCD – Pseudocode

```
Euclid(a, b)
  Input:  integers a, b  where a ≥ b ≥ 0
  Output: GCD(a, b)

  if b == 0
      return a
  return Euclid(b, a mod b)
```

Trace: GCD(24, 14)

Call	a	b	a mod b
1	24	14	10
2	14	10	4
3	10	4	2
4	4	2	0

GCD Implementation – Iteration vs Recursion

Iterative version:

```
def gcd_iter(a, b):
    while b ≠ 0:
        a, b = b, a % b
    return a
```

Recursive version:

```
def gcd_rec(a, b):
    if b == 0:
        return a
    return gcd_rec(b, a % b)
```

- Both produce identical results
- Recursion uses the **call stack**; each call pushes a new frame
- Iteration avoids stack overhead

Recursion and the Call Stack

```
gcd_rec(24, 14)
  └→ gcd_rec(14, 10)
        └→ gcd_rec(10, 4)
              └→ gcd_rec(4, 2)
                    └→ gcd_rec(2, 0) → returns 2
                 returns 2
            returns 2
      returns 2
returns 2
```

Each recursive call **pushes** a frame onto the stack. When `b == 0`, values **pop** back up.

Algorithm Representation

An algorithm is a step-by-step procedure — similar to a recipe.

Three common ways to express an algorithm:

Method	Description
Natural Language	Plain verbal description of each step
Pseudocode	Structured, language-like notation (most common)
Flowchart	Visual diagram with shapes and arrows

Example: Find the Maximum – Natural Language

Problem: Given n cards with numbers, find the largest number.

1. Read the first card's number and remember it
2. Read the next card and compare it with the remembered number
3. Keep the larger number in memory
4. If cards remain, go to step 2
5. The remembered number is the maximum

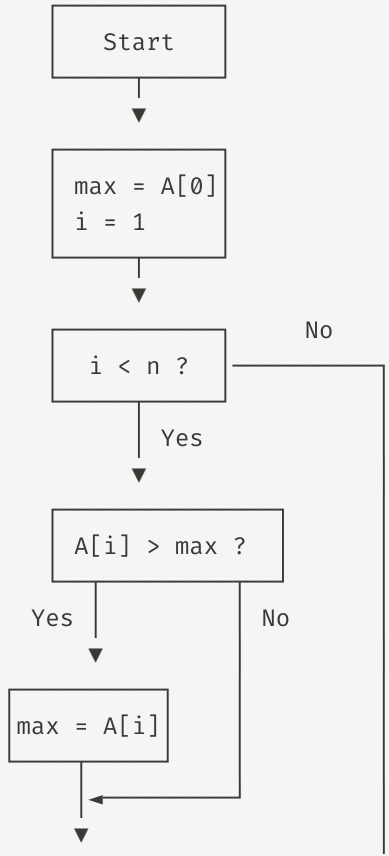
> Plain verbal description – easy to understand, but can be ambiguous.

Example: Find the Maximum – Pseudocode

```
FindMax(A[], n)
  max = A[0]
  for i = 1 to n-1
      if A[i] > max
          max = A[i]
  return max
```

- Structured notation – precise and unambiguous
- The most common representation in algorithm textbooks

Example: Find the Maximum – Flowchart



Algorithm Classification

Six major types based on problem-solving strategy

Classification by Problem-Solving Strategy

Type	Core Idea
Divide and Conquer	Split into subproblems → solve each → combine results
Greedy	Make the locally optimal choice at each step
Dynamic Programming	Solve subproblems, store results, reuse them
Approximation	Find near-optimal solutions for NP-hard problems
Backtracking	Explore all possibilities, prune infeasible branches
Branch and Bound	Like backtracking, but use bounds to prune more aggressively

Divide and Conquer

Idea: Break a large problem into smaller subproblems, solve each recursively, then combine.

Examples: Merge Sort, Quick Sort, Binary Search

```
          Problem(n)
         /          \
   Sub(n/2)      Sub(n/2)
     /  \          /  \
  S(n/4) S(n/4) S(n/4) S(n/4)
```

Strengths: Effective when the problem structure is naturally recursive.

Greedy Algorithms

Idea: At each step, choose the option that looks best **right now**.

- Does **not** always guarantee a globally optimal solution
- But under certain conditions (greedy-choice property + optimal substructure), it does

Examples:

- Minimum Spanning Tree (Kruskal's, Prim's)
- Dijkstra's Shortest Path
- Huffman Coding

Dynamic Programming (DP)

Idea: Break the problem into subproblems (like DaC), but **store** and **reuse** overlapping subproblem results.

Key conditions:

- Overlapping subproblems
- Optimal substructure

Examples:

- Fibonacci sequence
- Longest Common Subsequence (LCS)
- Knapsack problem

DP vs Divide and Conquer: DaC subproblems are independent; DP subproblems overlap.

Approximation, Backtracking, Branch-and-Bound

Approximation Algorithms

- For NP-hard problems where exact solutions are too expensive
- Trade optimality for speed; analyze the **approximation ratio**
- *Examples:* TSP approximation, Vertex Cover

Backtracking

- Exhaustive search with **pruning** of infeasible branches
- *Examples:* N-Queens, maze solving, permutation generation

Branch and Bound

- Like backtracking but computes **upper/lower bounds** to prune more effectively
- Used for optimization versions of NP-hard problems
- *Examples:* TSP (exact), Knapsack (exact)

Other Classifications

By problem domain:

- Sorting algorithms
- Graph algorithms
- Computational geometry

By computing environment:

- Parallel algorithms
- Distributed algorithms
- Quantum algorithms

Other:

- Artificial intelligence / Machine learning algorithms

Time Complexity

Measuring algorithm efficiency

What is Time Complexity?

- **Time complexity** expresses the number of **elementary operations** as a function of input size n
- **Elementary operations**: comparisons, reads, writes, arithmetic – simple constant-time operations

Example: Finding the maximum in n cards

- Sequential scan: compare each card with the current max
- Number of comparisons: $n - 1$
- Time complexity: $T(n) = n - 1$

Running Time Examples

Example 1 – O(1):

```
sample1(A[], n)
    k = n / 2
    return A[k]
```

One division + one array access → **constant time** regardless of n.

Running Time Examples

Example 2 – O(n):

```
sample2(A[], n)
    sum = 0
    for i = 1 to n
        sum = sum + A[i]
    return sum
```

The loop runs n times → **linear time**.

Running Time Examples

Example 3 – O(n^2):

```
sample3(A[], n)
    sum = 0
    for i = 1 to n
        for j = 1 to n
            sum = sum + A[i] * A[j]
    return sum
```

Nested loops, each running n times → $n \times n$ = **quadratic time**.

Running Time Examples

Example 4 – O(n^2) (triangular loop):

```
sample5(A[], n)
    sum = 0
    for i = 1 to n-1
        for j = i+1 to n
            sum = sum + A[i] * A[j]
    return sum
```

Inner loop runs $(n-1)+(n-2)+\cdots+1=\frac{n(n-1)}{2}$ times → still $O(n^2)$.

Running Time Examples

Example 5 – O(n) (recursion):

```
factorial(n)
    if n == 1 return 1
    return n * factorial(n - 1)
```

Recursion depth is n → **linear time**.

Types of Complexity Analysis

Type	Meaning
Worst-case	Upper bound – "no matter what input, it won't exceed this"
Average-case	Expected time under a probability distribution (usually uniform)
Best-case	Fastest possible execution – used to find optimal algorithms
Amortized	Average cost per operation over a sequence of operations

In practice, **worst-case analysis** is the most commonly used.

Commute Time Analogy

Scenario: Home → Station (6 min) → Subway (20 min) → Classroom (10 min)

Case	Time	Explanation
Best	36 min	Arrive at station, train is there immediately
Worst	40 min	Just missed the train, wait 4 min for the next one
Average	38 min	Average wait of ~2 min

This is how we think about algorithm analysis – the **worst case** gives us a guarantee.

Part 2. Asymptotic Notation

Why Asymptotic Notation?

- Time complexity is a function of n (usually a polynomial with multiple terms)
- As n grows large, only the **dominant term** matters
- **Asymptotic notation** simplifies the expression by focusing on growth rate

Three main notations:

Notation	Meaning	Intuition
O (Big-O)	Upper bound	Worst case — "at most this fast"
Ω (Big-Omega)	Lower bound	Best case — "at least this fast"
Θ (Theta)	Tight bound	Exact growth rate

Big-O Notation – Formal Definition

$O(g(n))$ = the set of functions that grow **at most as fast as** $g(n)$.

$$O(g(n)) = \{f(n) \mid \exists\, c > 0,\ n_0 \geq 0 \text{ s.t. } \forall\, n \geq n_0,\ f(n) \leq c \cdot g(n)\}$$

- $g(n)$ is the **asymptotic upper bound** of $f(n)$
- Convention: we write $f(n) = O(g(n))$ even though technically $f(n) \in O(g(n))$

Intuitive meaning:

$f(n) = O(g(n))$ means f does **not grow faster** than g (ignoring constant factors).

Big-O – Example Proof

Prove: $f(n) = 2n^2 - 8n + 3 = O(n^2)$

We need to find constants $c > 0$ and $n_0 \geq 0$ such that $2n^2 - 8n + 3 \leq c \cdot n^2$ for all $n \geq n_0$.

Choose $c = 5$:

$$2n^2 - 8n + 3 \leq 5n^2 \quad \text{for all } n \geq 1$$

This simplifies to $-3n^2 - 8n + 3 \leq 0$, which is true for $n \geq 1$.

Therefore, with $c = 5$ and $n_0 = 1$, we have $f(n) = O(n^2)$. $\square$

Members of O(n^2)

Functions in $O(n^2)$:

- $3n^2 + 2n$ ✓
- $7n^2 - 100n$ ✓
- $n \log n + 5n$ ✓
- $3n$ ✓

Functions NOT in $O(n^2)$:

- n^3 ×
- 3^n ×

Best practice: Be as **tight** as possible.

- $n \log n + 5n = O(n \log n)$ – do not write $O(n^2)$ even though technically correct
- A looser bound means **loss of information**

More Big-O Examples

Expression	Big-O	Why
$3n + 2$	$O(n)$	$3n + 2 \leq 4n$ for $n \geq 2$
$100n + 6$	$O(n)$	$100n + 6 \leq 101n$ for $n \geq 10$
$10n^2 + 4n + 2$	$O(n^2)$	$10n^2 + 4n + 2 \leq 11n^2$ for $n \geq 5$
$6 \cdot 2^n + n^2$	$O(2^n)$	$6 \cdot 2^n + n^2 \leq 7 \cdot 2^n$ for $n \geq 4$

Big-Omega Notation – Formal Definition

$\Omega(g(n))$ = the set of functions that grow **at least as fast as** $g(n)$.

$$\Omega(g(n)) = \{f(n) \mid \exists\, c > 0,\ n_0 \geq 0 \text{ s.t. } \forall\, n \geq n_0,\ f(n) \geq c \cdot g(n)\}$$

- $g(n)$ is the **asymptotic lower bound** of $f(n)$
- Symmetric to Big-O

Intuitive meaning:

$f(n) = \Omega(g(n))$ means f does **not grow slower** than g.

Big-Omega – Examples

Expression	Big-Omega	Why
$3n + 2$	$\Omega(n)$	$3n + 2 \geq 3n$ for $n \geq 1$
$100n + 6$	$\Omega(n)$	$100n + 6 \geq 100n$ for $n \geq 1$
$10n^2 + 4n + 2$	$\Omega(n^2)$	$10n^2 + 4n + 2 \geq n^2$ for $n \geq 1$
$6 \cdot 2^n + n^2$	$\Omega(2^n)$	$6 \cdot 2^n + n^2 \geq 2^n$ for $n \geq 1$

Best practice: Be as tight as possible.

- $n \log n + 5n = \Omega(n \log n)$ – don't write $\Omega(n)$

Theta Notation – Formal Definition

$\Theta(g(n))$ = the set of functions that grow **at the same rate as** $g(n)$.

$$\Theta(g(n)) = O(g(n)) \ \cap \ \Omega(g(n))$$

$$\Theta(g(n)) = \{f(n) \mid \exists\, c_1, c_2 > 0,\ n_0 \geq 0 \text{ s.t. } \forall\, n \geq n_0,\ c_2 \cdot g(n) \leq f(n) \leq c_1 \cdot g(n)\}$$

Intuitive meaning:

$f(n) = \Theta(g(n))$ means f and g grow at the **same rate** (up to constant factors).

Theta – Examples

Expression	Theta
$3n + 2$	$\Theta(n)$ – since $3n \leq 3n + 2 \leq 4n$ for $n \geq 2$
$10n^2 + 4n + 2$	$\Theta(n^2)$
$6 \cdot 2^n + n^2$	$\Theta(2^n)$

Non-examples:

$2n^2 + 3n + 5 \neq \Theta(n^3)$ and $2n^2 + 3n + 5 \neq \Theta(n)$

Theta requires **both** upper and lower bounds to match.

Common Complexity Classes

Notation	Name	Example
$O(1)$	Constant	Array index access
$O(\log n)$	Logarithmic	Binary search
$O(n)$	Linear	Linear search
$O(n \log n)$	Log-linear	Merge sort
$O(n^2)$	Quadratic	Bubble sort
$O(n^3)$	Cubic	Matrix multiplication (naive)
$O(2^n)$	Exponential	Brute-force subsets

Growth Rate Comparison

n	$\log n$	n	$n \log n$	n^2	n^3	2^n
10	3.3	10	33	100	1,000	1,024
100	6.6	100	664	10,000	10^6	10^{30}
1,000	10	1,000	10,000	10^6	10^9	10^{301}
10^6	20	10^6	2×10^7	10^{12}	10^{18}	–

The gap between polynomial and exponential is **enormous**.

Why Efficient Algorithms Matter

Sorting 1 billion numbers:

Algorithm	n = 1,000	n = 1,000,000	n = 1,000,000,000
$O(n^2)$ on PC	< 1 sec	~2 hours	**~300 years**
$O(n^2)$ on supercomputer	< 1 sec	~1 sec	~1 week
$O(n \log n)$ on PC	< 1 sec	< 1 sec	**~5 minutes**
$O(n \log n)$ on supercomputer	< 1 sec	< 1 sec	< 1 sec

An efficient algorithm is **more valuable than a supercomputer**.

Investing in better algorithms is far more economical than investing in better hardware.

Recurrence Relations

What is a Recurrence Relation?

A **recurrence relation** expresses a function in terms of its value on **smaller inputs**.

Examples:

- $a_n = a_{n-1} + 2$
- $f(n) = n \cdot f(n-1)$ – factorial
- $f(n) = f(n-1) + f(n-2)$ – Fibonacci
- $f(n) = f(n/2) + n$ – binary reduction

Why it matters: Many recursive algorithms naturally produce recurrence relations for their running time.

Merge Sort Recurrence

Merge Sort splits the array in half, sorts each half, then merges.

$$T(n) = 2T(n/2) + n$$

- $2T(n/2)$: two recursive calls on halves
- n: merging overhead (linear scan)
- $T(1) = 1$

Three Methods for Solving Recurrences

Method	Idea
Repeated Substitution	Expand the recurrence step by step until a pattern emerges
Guess and Verify	Guess the form, then prove by mathematical induction
Master Theorem	Apply a formula directly to recurrences of a specific form

Method 1: Repeated Substitution

Example: $T(n) = T(n-1) + c, T(1) \leq c$

$$
\begin{aligned}
T(n) &= T(n-1) + c \\
&= (T(n-2) + c) + c = T(n-2) + 2c \\
&= T(n-3) + 3c \\
&\vdots \\
&= T(1) + (n-1)c \\
&\leq c + (n-1)c = cn = O(n)
\end{aligned}
$$

This is the recurrence for **factorial** – linear time.

Repeated Substitution – Merge Sort

Recurrence: $T(n) = 2T(n/2) + n, T(1) = 1$

Assume $n = 2^k$:

$$
\begin{aligned}
T(n) &= 2T(n/2) + n \\
&= 2(2T(n/4) + n/2) + n = 4T(n/4) + 2n \\
&= 8T(n/8) + 3n \\
&\vdots \\
&= 2^k T(n/2^k) + kn \\
&= n \cdot T(1) + n \log_2 n \\
&= n + n \log n = O(n \log n)
\end{aligned}
$$

Method 2: Guess and Verify (Induction)

Recurrence: $T(n) = 2T(n/2) + n$

Guess: $T(n) = O(n \log n)$, i.e., $T(n) \leq cn \log n$ for some constant $c > 0$.

Proof by induction:

Base case: $T(2) \leq c \cdot 2 \log 2$ – holds for sufficiently large c.

Inductive hypothesis: Assume $T(n/2) \leq c(n/2) \log(n/2)$.

Inductive step:

$$\begin{aligned} T(n) &= 2T(n/2) + n \\ &\leq 2 \cdot c(n/2) \log(n/2) + n \\ &= cn \log n - cn \log 2 + n \\ &= cn \log n + (-c \log 2 + 1)n \\ &\leq cn \log n \quad (\text{when } c \geq 1/\log 2) \end{aligned}$$

Therefore $T(n) = O(n \log n)$. $\square$

Guess and Verify – Common Pitfall

Caution: The constant c in the hypothesis must be the **same** c in the conclusion.

Wrong proof attempt for $T(n) = 2T(n/2) + 1$, guess $T(n) = O(n)$:

$$T(n) \leq 2 \cdot c(n/2) + 1 = cn + 1$$

This is $cn + 1$, **not** $\leq cn$. The proof fails!

Fix: Strengthen the guess to $T(n) \leq cn - 2$:

$$T(n) \leq 2(c(n/2) - 2) + 1 = cn - 4 + 1 = cn - 3 \leq cn - 2 \checkmark$$

Master Theorem

Master Theorem – Statement

For recurrences of the form:

$$T(n) = aT(n/b) + f(n) \quad \text{where } a \geq 1,\ b > 1$$

Let $h(n) = n^{\log_b a}$. Compare $f(n)$ with $h(n)$:

Case	Condition	Result
Case 1	$f(n) = O(n^{\log_b a - \varepsilon})$ for some $\varepsilon > 0$	$T(n) = \Theta(n^{\log_b a})$
Case 2	$f(n) = \Theta(n^{\log_b a})$	$T(n) = \Theta(n^{\log_b a} \log n)$
Case 3	$f(n) = \Omega(n^{\log_b a + \varepsilon})$ for some $\varepsilon > 0$, and $af(n/b) \leq cf(n)$ for some $c < 1$	$T(n) = \Theta(f(n))$

Master Theorem – Intuition

$$T(n) = aT(n/b) + f(n)$$

The total work is: $f(n) + a \cdot f(n/b) + a^2 \cdot f(n/b^2) + \cdots + n^{\log_b a} \cdot T(1)$

Who dominates?

Scenario	Winner	Result
$h(n) = n^{\log_b a}$ dominates $f(n)$	Leaf work wins	$T(n) = \Theta(h(n))$
$f(n)$ dominates $h(n)$	Root work wins	$T(n) = \Theta(f(n))$
$h(n)$ and $f(n)$ are equal	Balanced – each level contributes equally	$T(n) = \Theta(h(n) \log n)$

Master Theorem – Examples

Example 1: $T(n) = 2T(n/3) + c$

- $a = 2, b = 3, h(n) = n^{\log_3 2} \approx n^{0.63}, f(n) = c = O(1)$
- $f(n)$ is polynomially smaller → **Case 1**
- $T(n) = \Theta(n^{\log_3 2})$

Example 2: $T(n) = 2T(n/4) + n$

- $a = 2, b = 4, h(n) = n^{\log_4 2} = n^{0.5} = \sqrt{n}, f(n) = n$
- $f(n)$ is polynomially larger → **Case 3**
- $T(n) = \Theta(n)$

Example 3: $T(n) = 2T(n/2) + n$

- $a = 2, b = 2, h(n) = n^{\log_2 2} = n, f(n) = n$
- $f(n) = \Theta(h(n))$ → **Case 2**
- $T(n) = \Theta(n \log n)$

Summary

- **Algorithm** = step-by-step problem-solving procedure (correctness, executability, finiteness, efficiency)
- **Representation**: pseudocode is the standard; also natural language and flowcharts
- **Classification**: DaC, Greedy, DP, Approximation, Backtracking, Branch-and-Bound
- **Time Complexity**: count elementary operations as a function of input size n
- **Asymptotic Notation**: O (upper bound), Ω (lower bound), Θ (tight bound)
- **Recurrence Relations**: repeated substitution, guess-and-verify, Master Theorem
- **Textbook**: CLRS Chapters 1–3

Homework

- **Assignment 1** is posted – check LMS for details
- No quiz this week; quizzes begin **Week 03**
- Next week: **Arrays, Sorting Algorithms** – elementary and advanced sorts

Q & A

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